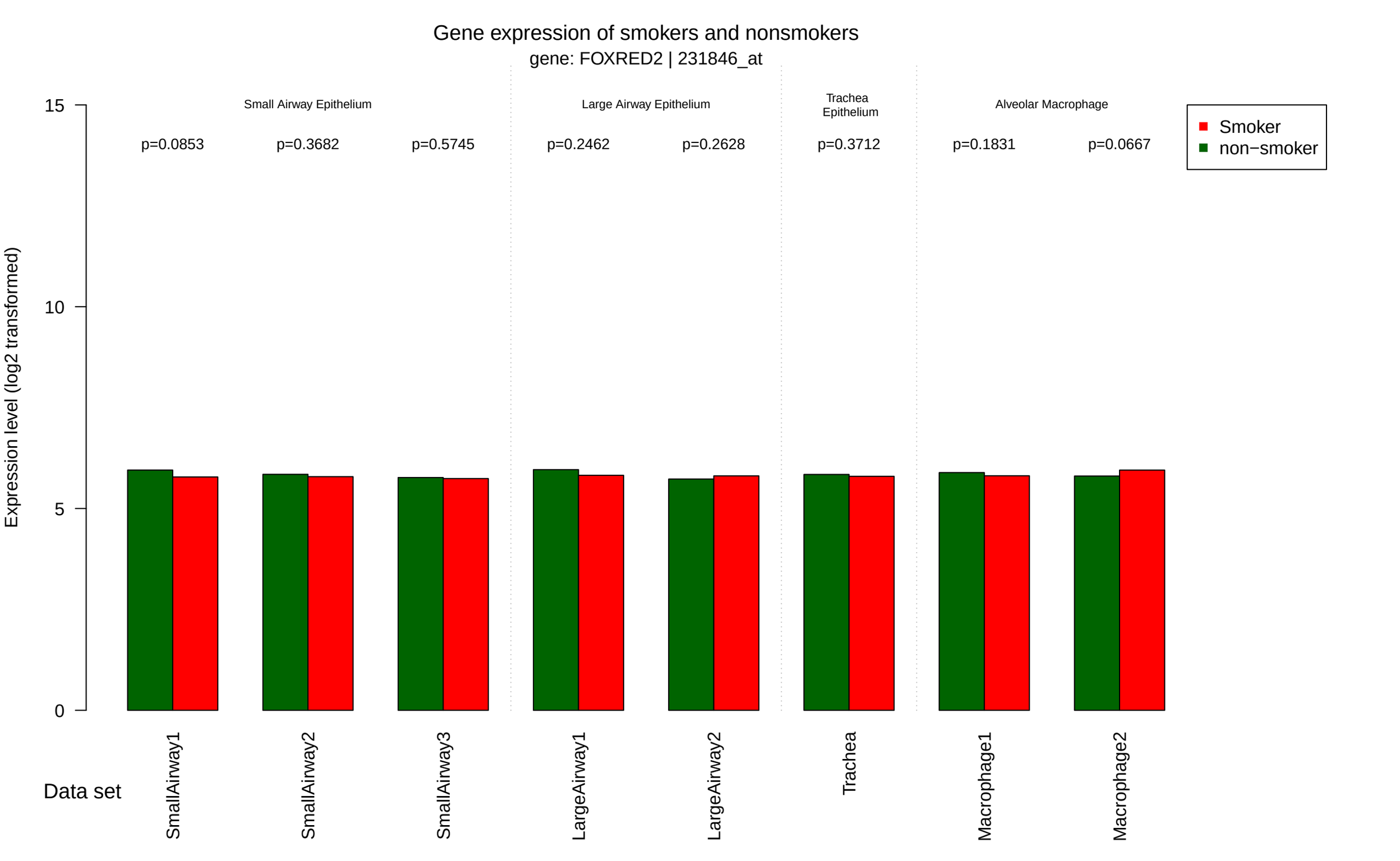
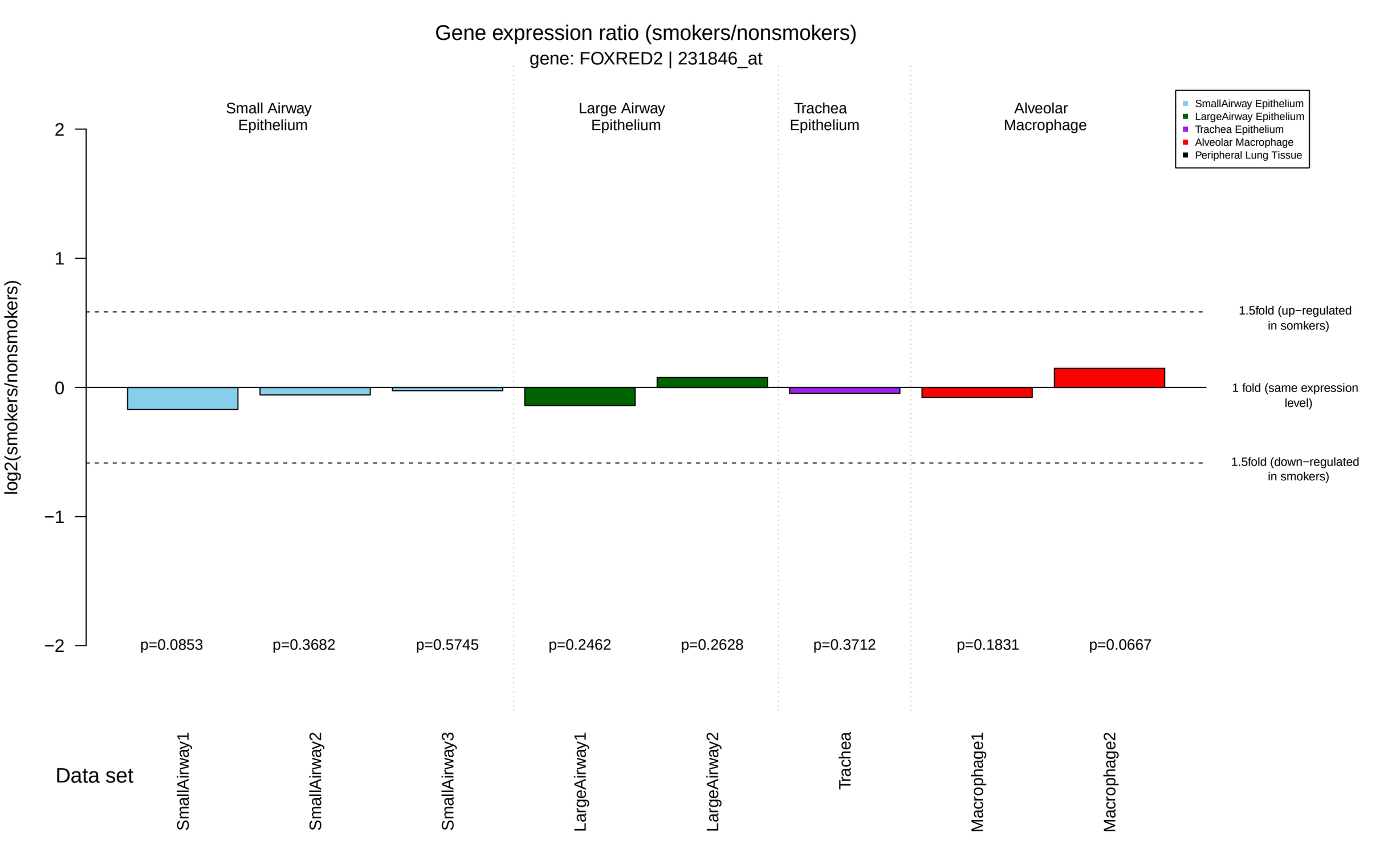
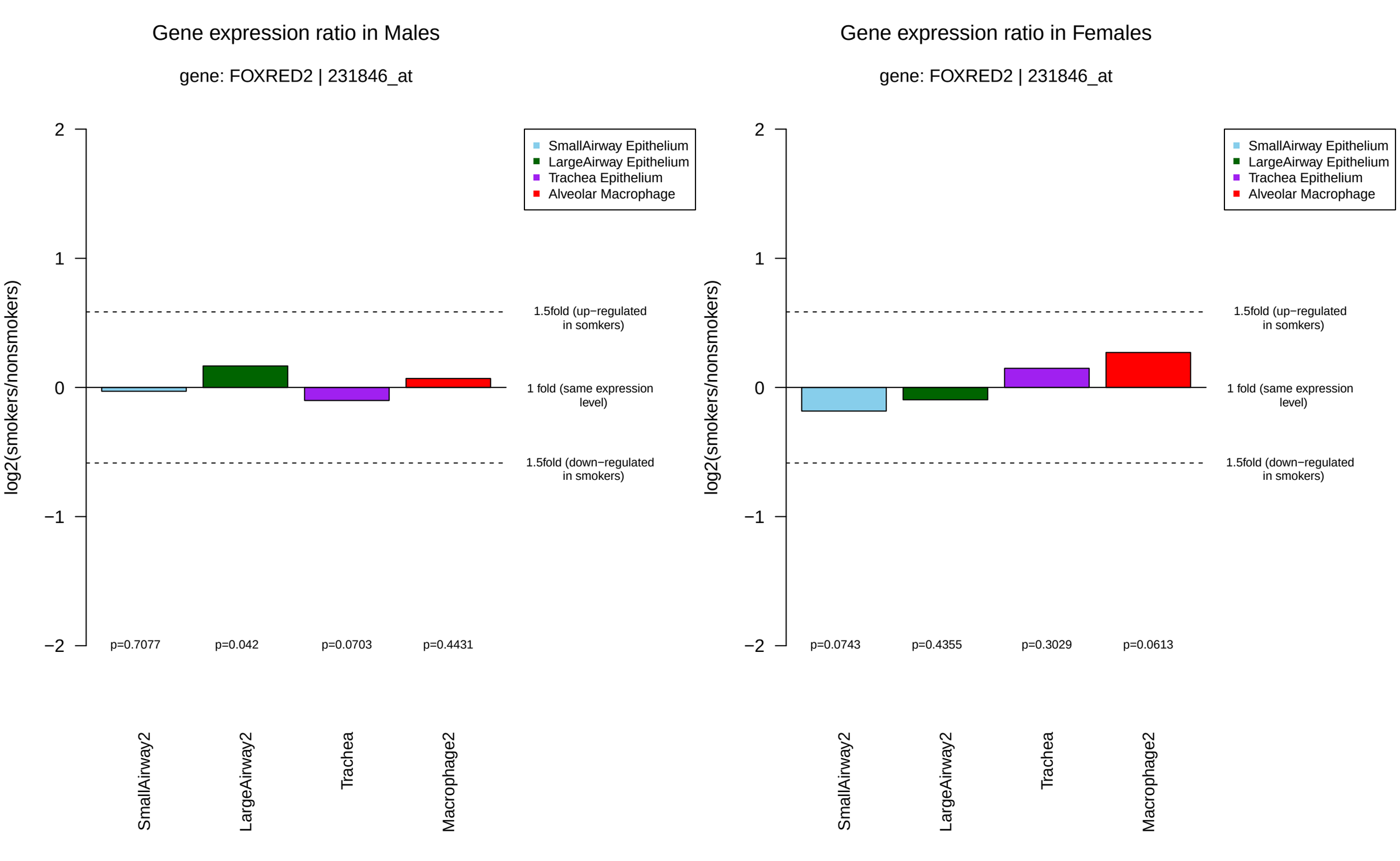
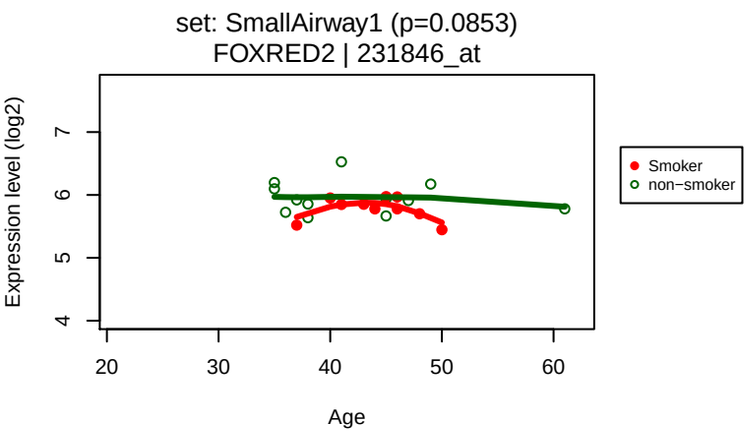
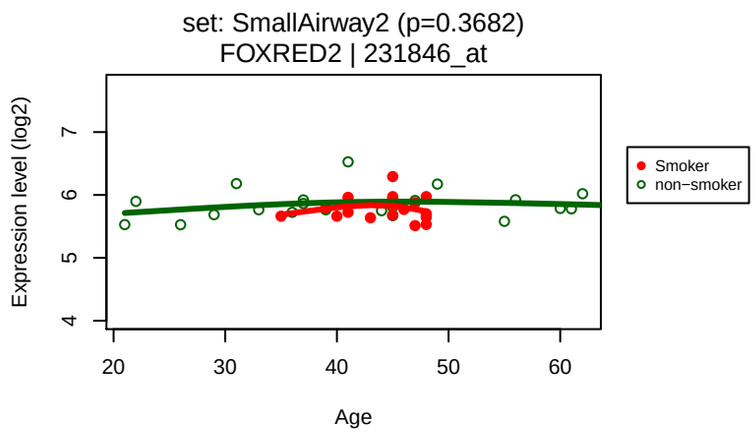
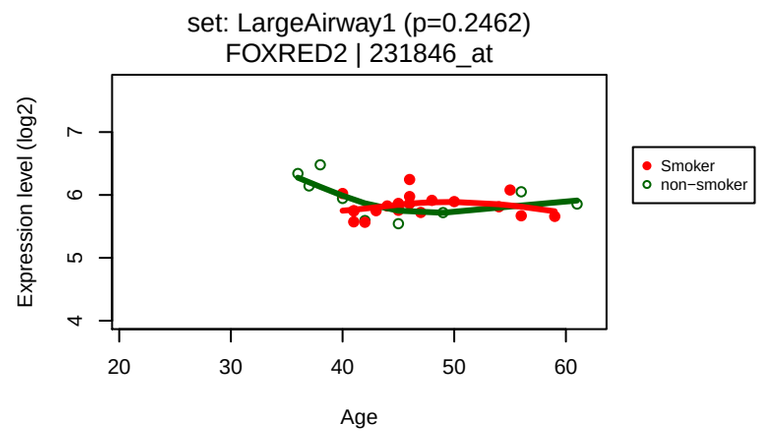

Gene expression of smokers and nonsmokers
gene: FOXRED2 | 231846_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0853
p=0.3682
p=0.5745
p=0.2462
p=0.2628
p=0.3712
p=0.1831
p=0.0667
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

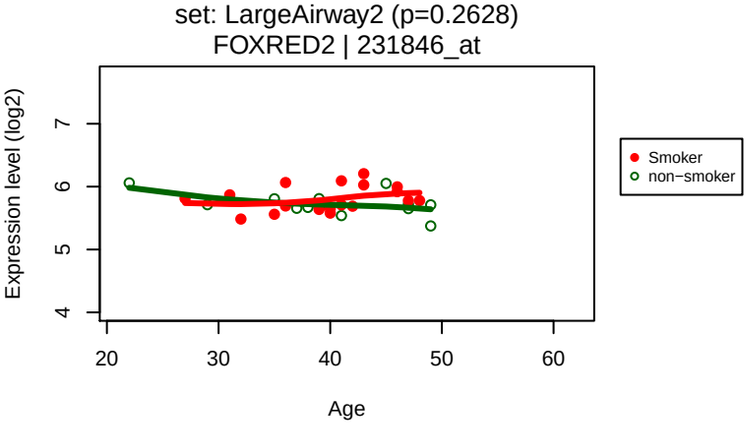

Gene expression ratio (smokers/nonsmokers)
gene: FOXRED2 | 231846_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0853
p=0.3682
p=0.5745
p=0.2462
p=0.2628
p=0.3712
p=0.1831
p=0.0667
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

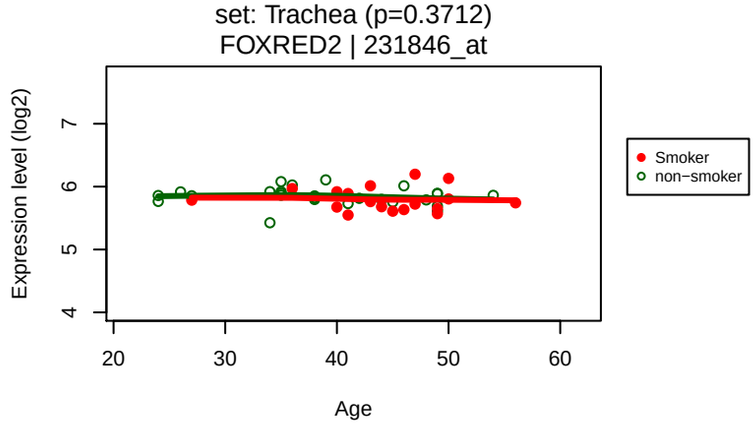

Gene expression ratio in Males
gene: FOXRED2 | 231846_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7077
p=0.042
p=0.0703
p=0.4431
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FOXRED2 | 231846_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0743
p=0.4355
p=0.3029
p=0.0613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

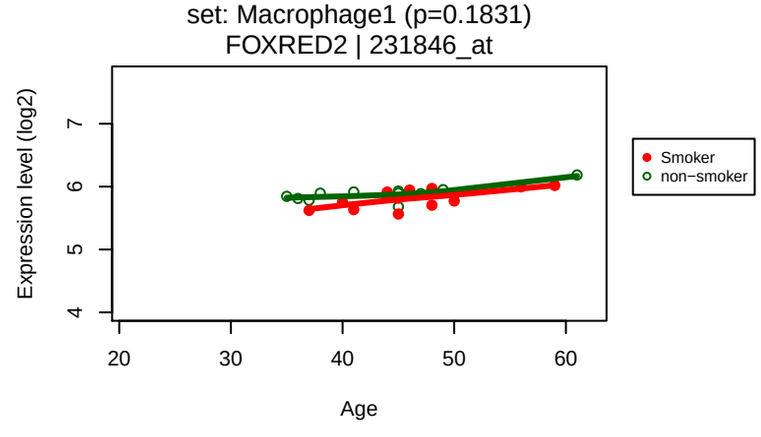

set: SmallAirway1 (p=0.0853)
FOXRED2 | 231846_at
Expression level (log2)
Age
Smoker
non−smoker

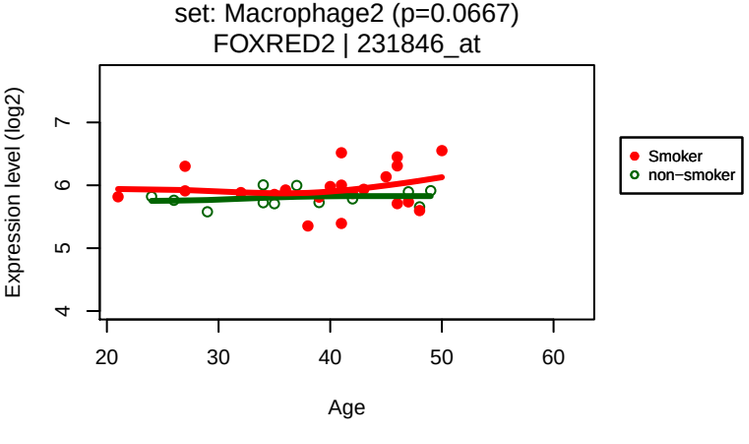

set: SmallAirway2 (p=0.3682)
FOXRED2 | 231846_at
Expression level (log2)
Age
Smoker
non−smoker

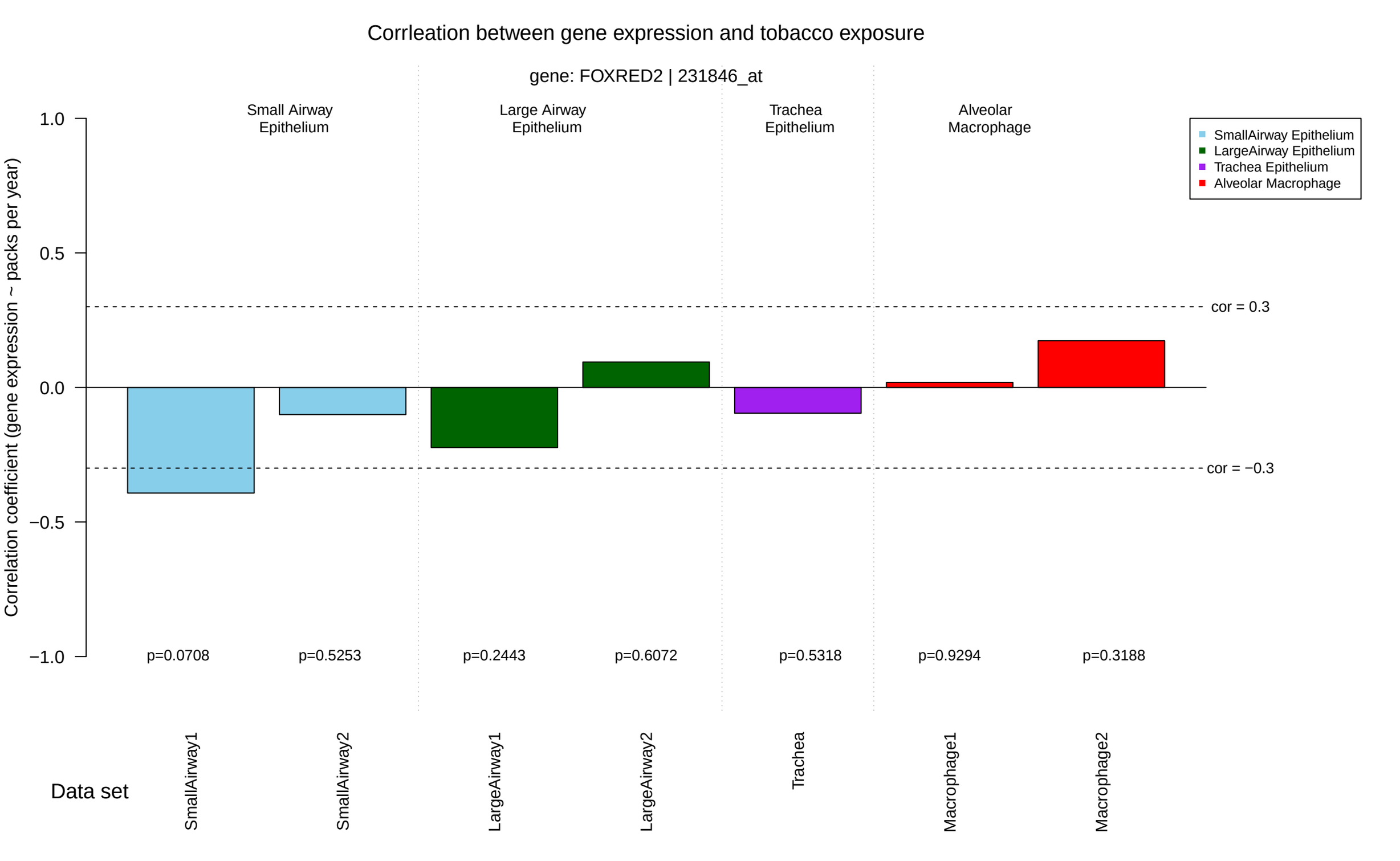

set: LargeAirway1 (p=0.2462)
FOXRED2 | 231846_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2628)
FOXRED2 | 231846_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3712)
FOXRED2 | 231846_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1831)
FOXRED2 | 231846_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0667)
FOXRED2 | 231846_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FOXRED2 | 231846_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0708
p=0.5253
p=0.2443
p=0.6072
p=0.5318
p=0.9294
p=0.3188
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2